

15 Story-Based Coding Challenges

For Loops, If-Else, Dictionaries, Lists & Tuples — Easy to Medium

Each problem is dressed up as a short story, in the style of LeetCode/CodeChef-flavored practice questions. Core concepts tested: for loops, if-else, dictionaries, lists, and tuples. Full solutions — with code and line-by-line explanations — are in the companion document.

1. Candy Store Gift Card (Easy — Dictionary)

Riya is at a candy store with a gift card worth exactly `target` rupees. Each candy in the display has a price tag, stored in the list `nums`. She wants to pick exactly two candies whose prices add up to exactly the gift card's value. Help her find the *positions* of those two candies on the shelf.

Input: `nums = [2,7,11,15]`, `target = 9` → Output: `[0,1]`

2. The Suspicious Attendance Register (Easy — List + Dictionary)

A teacher is collecting roll numbers called out during attendance. She suspects a mischievous student answered twice for a friend who's absent. Given the list of roll numbers called out, tell her whether any roll number was called more than once.

Input: `[1,2,3,1]` → Output: `True`

3. The Scrambled Name Game (Easy — Dictionary)

Two friends, Aman and Tara, are playing a word game where they scramble each other's names to form new words. Aman claims the word he wrote, `t`, is just a scrambled version of Tara's original word, `s`. Verify whether Aman is telling the truth.

Input: `s = "anagram"`, `t = "nagaram"` → Output: `True`

4. The Village Election (Easy — Dictionary)

In a small village election, every voter shouts out the name of their preferred candidate (represented as numbers in the array). The village has a rule: a candidate only wins outright if they got votes from **more than half** the villagers. Find who that candidate is.

Input: `[2,2,1,1,1,2,2]` → Output: `2`

5. The Joint Reading Event (Easy — List + Dictionary)

Two libraries, `nums1` and `nums2`, each maintain a list of book ID numbers they own. The city wants to know which book IDs exist in **both** libraries' collections, so it can plan a joint reading event — listing each shared book ID only once.

Input: `nums1 = [1,2,2,1]`, `nums2 = [2,2]` → Output: `[2]`

6. The Detective's Ransom Note (Easy — Dictionary)

A detective is examining a ransom note (a string of letters) for clues. He believes the very first letter that appears **only once** in the note is secretly the culprit's initial. Find the index of that letter — or tell him no such letter exists.

```
Input: "leetcode" → Output: 0
```

7. The Archaeologist's Tablet (Easy — Dictionary + If-Else)

An archaeologist has uncovered an ancient Roman tablet inscribed with a number in Roman numerals. Translate the inscription into a regular integer so the museum can catalog it correctly — remembering that the Romans sometimes wrote a smaller numeral before a larger one to mean subtraction (like "IV" for 4).

```
Input: "MCMXCIV" → Output: 1994
```

8. The Tribal Chant (Easy/Medium — Dictionary)

A language researcher is studying a tribe's chant, where each word is believed to consistently represent one specific animal from a known pattern (like "abba"). Given the pattern and the actual chant (a string of words), determine if the chant truly follows that pattern consistently, with no word representing two different letters or vice versa.

```
Input: pattern = "abba", s = "dog cat cat dog" → Output: True
```

9. The Cashier's Coupon Stack (Easy/Medium — List + Tuple-style pairing)

A cashier has a **sorted** stack of coupons, each with a printed value. A customer wants to combine exactly two coupons to cover a bill of exactly `target` rupees. Since coupons are numbered starting from 1 (not 0) in the stack, tell the cashier the **positions** (as a pair) of the two coupons to hand over.

```
Input: numbers = [2,7,11,15], target = 9 → Output: (1,2)
```

10. The Music App's Daily Highlight Reel (Medium — Dictionary + Sorting)

A music streaming app logs every song ID a user plays throughout the day. At midnight, the app wants to show the user their "Top K Most Played Songs" of the day as a highlight reel. Given the day's play log and a number `k`, find those top `k` song IDs.

```
Input: nums = [1,1,1,2,2,3], k = 2 → Output: [1,2]
```

11. The Librarian's Puzzle Shelf (Medium — Dictionary)

A librarian is reorganizing a shelf of word-puzzle cards, and wants to group together all cards that are scrambled versions of each other (anagrams), so puzzle-solvers can find related cards easily. Given the list of cards, organize them into these groups.

```
Input: ["eat", "tea", "tan", "ate", "nat", "bat"]  
Output: [["eat", "tea", "ate"], ["tan", "nat"], ["bat"]]
```

12. The Shopkeeper's Ledger (Medium — Dictionary + Prefix Sum)

A shopkeeper recorded his daily profits and losses (positive and negative numbers) over a stretch of days in the array `nums`. He wants to know, out of all the possible **consecutive stretches of days**, how many of them added up to exactly `k` rupees in total profit.

```
Input: nums = [1,1,1], k = 2 → Output: 2
```

13. The Warehouse Scanner Glitch (*Medium — List*)

A warehouse worker scanned a batch of parcels, where each parcel's ID should normally be scanned only once — but due to a barcode scanner glitch, some parcels got scanned exactly twice. Given the full scan log (where every ID appears once or twice), find all the parcel IDs that were mistakenly scanned twice.

```
Input: [4,3,2,7,8,2,3,1] → Output: [2,3]
```

14. The Coin Collector's Display (*Medium — Dictionary/Set*)

A collector has a messy box of antique coins, each stamped with a year, thrown in with no particular order. He wants to know the length of the **longest unbroken run of consecutive years** he owns (e.g., coins from 1,2,3,4 in a row), to see how close he is to completing a “consecutive years” display set.

```
Input: [100,4,200,1,3,2] → Output: 4
```

15. The Conference Hall Booking Clash (*Medium — List of Tuples + If-Else*)

A conference hall manager has a list of meeting time slots booked by different teams, each given as a `(start, end)` pair. Some bookings overlap by mistake. Help him merge all the overlapping time slots into single, combined bookings so the hall's calendar is accurate.

```
Input: [(1,3),(2,6),(8,10),(15,18)]  
Output: [(1,6),(8,10),(15,18)]
```

Solutions — with code and a full line-by-line explanation for each of the 15 stories above — are provided in the companion solutions document.